LECD: Label-Efficient Completion Detection for Ballet Rotation Evaluation

Jiuhui Xu
High school student
The High School Affiliated to Renmin University of China
Beijing, China
jiuhuixu29@gmail.com

Zhimin Lin
The High School Affiliated to Renmin University of China
Beijing, China
linzhimin@rdfz.cn

Sisi Wang
The High School Affiliated to Renmin University of China
Beijing, China
wangsisi@rdfz.cn

***Abstract*—Action quality assessment (AQA) usually assumes a localized and complete action, whereas ballet videos may contain several dancers or show an incomplete attempt. We propose a Label-Efficient Completion Detection (LECD), which checks dancer-specific completion, eligibility, and visibility before estimating quality. LECD-Completion (LECD-C) fuses 18 observed 2D pose-dynamics descriptors with 11 rotation-physics descriptors from a frozen MotionBERT lift. On the locked test, it achieves 0.917 balanced accuracy, versus 0.833 balanced accuracy for the strongest fixed pose-only baseline. A separate quality head LECD-Quality (LECD-Q) obtains 0.431 Macro-F1 and 1.364 MAE on 43 expert-scored development events, which has the best performance among baselines. Evidence shows that the model can be used on videos to get completion detection before quality assessment to provide dancers a more reliable data-based analysis of their movements and further identify key research direction of refined rotation quality assessment.**

Keywords—Ballet rotation, Active learning, Machine Learning, MotionBERT, Expert labeling.

I. INTRODUCTION

AQA methods generally assume that the evaluated action is centered, temporally localized, and complete. Natural ballet footage violates this assumption when clips end before landing, merge adjacent turns, contain several dancers, or show attempts below one revolution. Scoring such clips conflates event validity with execution quality.

LECD separates these decisions. LECD-C first detects a complete, unassisted, visible dancer-specific turn by fusing observed 2D pose dynamics with frozen-lift rotation evidence. Only events that pass the gates enter the exploratory ordinal scorer LECD-Q.

A high-recall pose pipeline proposes 270 events from the 114-source [1-4] core corpus and 39 manually localized rotation events from 14 legacy sources [5-18]; ballet experts' response of boundary clarification and source-level filtering produces 95 development events from 74 sources and a prospective test of 12 unseen sources. Active sampling method is used when selecting part of development data which especially need experts' interpretation, and all those data are sent to Ballet teachers for labeling. Furthermore, final model and 12 test data with prospective probabilities are sealed before expert-label return is done, preventing post-label feature or threshold revision. On a label-blind prospective test, LECD-C reaches 0.917 balanced accuracy, achieved the highest rate among baselines.

II. RELATED WORK

AQA models predict absolute scores [19], distributions [20], relative quality [21], or procedure-aware structure [22, 23], but normally assumes a localized action. Instead, in LECD, assessing the quality on completion and treating boundary correction are parts of the protocol. Large RGB and human-motion models provide rich representations [24–27], but due to limited labels, MotionBERT [28] is used as a frozen 3D lift, training compact heads, and RGB is used to extract body skeleton. Active-learning methods combine uncertainty and coverage [29–33], so it can be used to select samples that are in doubt and difficult and should be given to expert first, which is a naive generalization estimates acquiring samples bias [34]. Therefore, the modeling restricts acquisition to development and evaluates once on separately locked sources rather than using those samples as the test set.

III. COMPLETION LABEL

For a source video $V = \{I_t\}_{t=1}^{T}$ and tracked dancer *d*, event X_d spans preparation state through landing or recovery and contains RGB frames, 2D keypoints, and confidences. Each rotating movement receives a separate record. The expert completion label is:

$$y_d^{\text{det}} \in \{0,1,2\} \tag{1}$$

Equation (1) defines the expert event label: 0 denotes no rotation, 1 an incomplete attempt below one continuous 360-degree revolution, and 2 at least one complete revolution. The binary detector maps labels 0 and 1 to noncomplete and retains label 2 as complete.

For complete, eligible, and scorable events, the expert assigns $q_d \in \{1,...,7\}$. For better learning, the primary target is

$$y_d^{\text{det}} = \begin{cases} 0, & q_d \in \{1,2,3\} \mid (\text{low}) \\ 1, & q_d \in \{4,5\} \mid (\text{medium}) \\ 2, & q_d \in \{6,7\} \mid (\text{high}) \end{cases} \tag{2}$$

where 1–7 score is auxiliary and suggestive.

IV. DATA AND EXPERT ANNOTATION

A. Data selection

The dataset contains 114 real-human videos: 83 UID-Ballet [1], 16 UCF-CIL [2], 14 DynamicPredictions stimuli [3], and one Wikimedia video [4]. In addition to the 114-source core candidate corpus, 39 manually localized rotation clips were collected from 14 legacy source videos [5-18]. During data collection, both high- and low-quality performances from professional and amateur dancers were

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included to ensure diversity in skill level and execution quality. A high-recall YOLO11s-Pose pipeline [35] tracked dancers and proposes intervals around changes in a shoulder–hip orientation proxy, yielding 270 proposals from 99 sources. Expert review then identified materials that have multiple dancers, so the source-level removal left 203 unassisted candidates for downstream modeling.

The 55-event expert core comprised 40 frozen anchors and 15 additions chosen by

$$A_i = 0.35U_i + 0.25D_i + 0.20N_i + 0.20E_i \qquad (3)$$

where uncertainty U, coverage distance D, source novelty N, and expected annotation efficiency E were percentile-normalized. Greedy distance updates and source caps limited redundancy. Fig. 1 shows how the 15 additions extend coverage of the frozen anchors in the proposal pool.

After review, the same representation proposed 15 follow-up events by balancing uncertainty and coverage, as shown in Fig. 2.

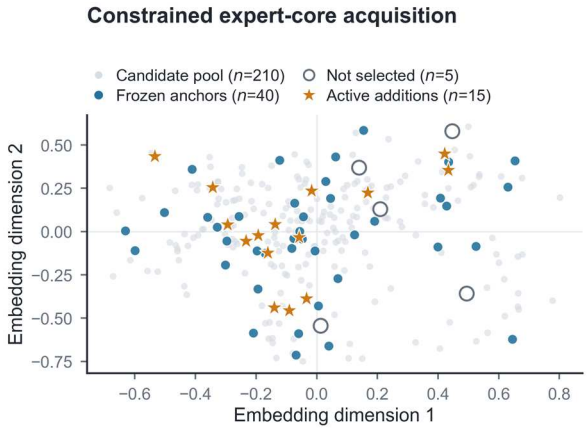


Fig. 1. Constrained expert-core acquisition, showing the frozen anchors and active additions.

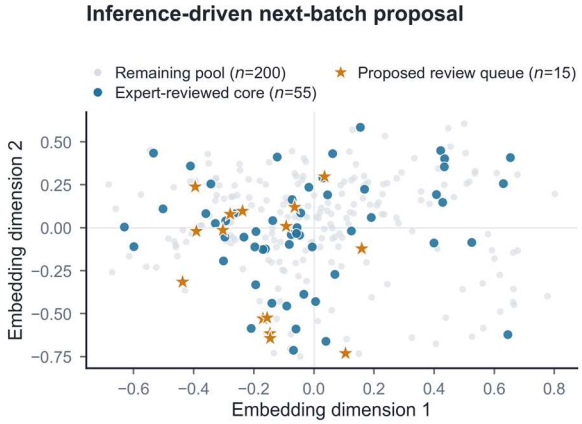


Fig. 2. Inference-driven next-batch proposal.

The actual sample selection was performed using the complete 24-dimensional feature representation. The orange stars indicate candidates for human review.

B. *Experts' labeling and construction of development events*

Experts received anonymized RGB, skeletons, evidence frames, and blank forms, but they were blinded to source identity, machine predictions, acquisition scores, and prior ratings to reduce anchoring bias. They recorded completion (0/1/2), eligibility, family, revolution count, visibility, and quality when valid. Because proposals may truncate or merge turns, experts relocalized one ordinary turn per event, and labels were not transferred across revised boundaries. According to the experts' return, turns in pas de deux (partnered dance) were removed by source. As shown in Table I, the frozen unassisted manifest contains 203 events. Experts' boundary review and labeling produced 95 development events from 74 sources (48 complete, 47 noncomplete) and 43 quality-scored events from 35 sources (8/16/19 low/medium/high). These samples were restricted to development.

TABLE I. EXPERT-GROUNDED DATA.

Layer	Events / records
Machine proposals / unassisted manifest	270 / 203
Initial / active / quality-supplement labels	44 / 21 / 18
Boundary-review labels	12
Detection development (74 sources)	95
Strict quality development (35 sources)	43
Locked test: complete / incomplete / none	6 / 0 / 6

C. *Construction of perspective labels (test events)*

Before label return, we fixed the features, model families, ensemble seeds, 0.8/0.2 fusion, and 0.5 threshold, then selected 12 target-centered clips from 12 unseen sources by diversity and pose-space coverage. Files were SHA-256 locked, source overlap with development was zero, and two blinded experts agreed on 11 of 12 clips for both ternary and binary completion labels.

V. LECD

Fig.3 summarizes the implemented pipeline from a normalized 64-frame pose sequence to selective assessments.

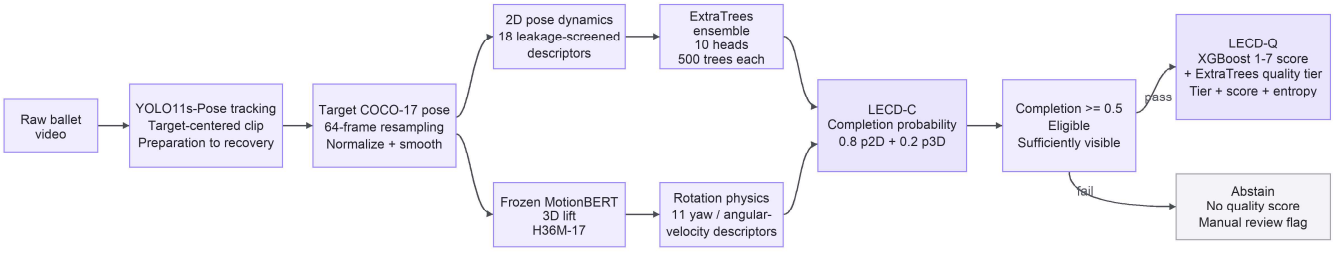


Fig. 3. System overview.

A. 2D dynamics

Each expert-repaired event is sampled at 64 relative times. YOLO11s-Pose selects the target COCO-17 track using anchor proximity, box scale, confidence, and temporal continuity. Coordinates are box-normalized, smoothed by a length-7 order-2 Savitzky–Golay filter, and interpolated to 64 times. Eighteen leakage-screened descriptors summarize orientation range, joint motion, symmetry, support-leg and torso stability, and pose confidence; ten class-balanced ExtraTrees heads produce p_i^{2D}. (4) centers every joint on the target box and scales it by the larger box dimension, reducing translation and scale variation before the 18 descriptors are computed. Fig. 4 shows the observed and lifted representations on active-rotation frames.

$$p_{ikj}^{\text{norm}} = \frac{2(p_{ikj} - c_{ik})}{s_{ik}}, j = 1, \dots, 17 \quad (4)$$

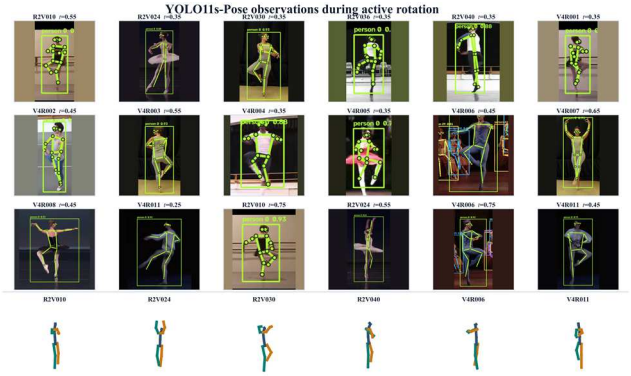


Fig. 4. Representative active-rotation observations.

B. Frozen 3D rotation representation and fusion

COCO-17 sequences are mapped to H36M-17 and lifted by frozen MotionBERT. Original and anatomically reflected lifts are unreflected and averaged. Shoulder and hip transverse vectors define torso orientation; cumulative angular path, central path, 90th-percentile step, and confidence statistics form 11 rotation descriptors. Ten class-balanced RandomForest heads produce p_i^{3D}. These values are rotation evidence, not calibrated yaw or revolution labels. (5) averages the original lift with the reflected lift after undoing the reflection, reducing left–right view sensitivity before the 11 rotation descriptors are computed.

$$Q_i = \frac{1}{2}\left[\mathcal{M}(P_i^{\text{H36M}}) + \mathcal{F}^{-1}\left(\mathcal{M}(\mathcal{F}(P_i^{\text{H36M}}))\right)\right] \quad (5)$$

$$p_i^{\text{comp}} = 0.8p_i^{2D} + 0.2p_i^{3D}, z_i^{\text{pred}} = \mathbb{1}[p_i^{\text{comp}} \geq 0.5] \quad (6)$$

Equation (6) is the frozen completion rule: it assigns weight 0.8 to the stronger development-time 2D probability and 0.2 to the complementary 3D probability, then predicts complete at 0.5. The weight maximized source-grouped development out-of-fold balanced accuracy; neither the weight nor the threshold used locked labels.

C. Selective quality

Only events passing completion and validity gates receive quality output. A regularized XGBoost regressor predicts the auxiliary 1–7 score and an induced low/medium/high distribution; class-balanced ExtraTrees predicts the three tiers directly. Their equal-weight probability average gives the released tier, and normalized entropy flags review. (7) averages the XGBoost-induced and ExtraTrees tier distributions; the argmax returns the released low/medium/high tier, while entropy remains a review flag.

$$\pi_i^Q = \frac{1}{2}\pi_i^{\text{XGB}} + \frac{1}{2}\pi_i^{\text{ET}}, q_i^{\text{pred}} = \text{argmax}_{k\in\{1,2,3\}}\pi_{ik}^Q \quad (7)$$

VI. Experimental Protocol

To prevent source leakage, all events from the same video were assigned to the same fold. Completion detection was trained on 95 development events using 10 repetitions of stratified, source-grouped five-fold cross-validation, with progressive model fitting repeated within each training fold. We got one consensus prediction per event by averaging repeated out-of-fold probabilities.

Baselines are majority prior, pose logistic regression, XGBoost, ExtraTrees, 3D physics, and a frozen MotionBERT embedding with fold-fitted standardization, PCA-8, and class-balanced logistic regression.

The locked 12-source test is evaluated once with frozen settings. The MotionBERT-embedding test row is explicitly retrospective because its probabilities were generated after label return; all other locked rows were sealed.

Metrics are balanced accuracy, F1, ROC AUC, and Brier score for completion, and Macro-F1, balanced accuracy, QWK, tier MAE, and 1–7 score MAE for quality. Quality uses 12 repeated source-grouped five-fold evaluations on 43 events, with ordinal thresholds fitted only on inner grouped predictions.

Implementation Details

The 2D branch averages ten class-balanced ExtraTrees models with 500 trees each; the 3D branch averages ten class-balanced RandomForest models with 600 trees each. Logistic regression uses median imputation, standardization, class weighting, and C = 0.1. XGBoost uses 100 depth-2 trees, learning rate 0.05, row and column subsampling 0.8, minimum child weight 3, L1=0.2, and L2=10. The fusion sweep uses development out-of-fold probabilities only. All ablations share folds, seeds, preprocessing, and the fixed 0.5 decision threshold; confidence intervals use paired source-cluster bootstrap resamples.

VII. Results

A. Completion detection

Table II reports completion results. LECD-C reaches 0.800 balanced accuracy on source-grouped development consensus, versus 0.747 for 2D ExtraTrees, 0.768 for 3D physics, and 0.664 for the frozen MotionBERT embedding. On the locked test it reaches 0.917 balanced accuracy and 0.972 AUC, with 6 true negatives, 5 true positives, 1 false negative, and no false positives as shown in Table III . XGBoost has the lower Brier score (0.135 versus 0.158).

TABLE II. COMPLETION DETECTION.

Model	Bal. acc.	F1	AUC	Brier
(a) Development consensus (n = 95)				
Majority prior	0.500	0.671	0.475	0.250
Pose logistic	0.673	0.705	0.735	0.207
XGBoost	0.684	0.694	0.756	0.206
ExtraTrees	0.747	0.760	0.789	0.204
3D physics	0.768	0.771	0.777	**0.191**
MotionBERT emb. + logit	0.664	0.652	0.729	0.212
LECD-C	**0.800**	**0.804**	**0.822**	**0.191**
(b) Locked prospective test (n = 12)				
Majority prior	0.500	0.667	0.500	0.250
Pose logistic	0.750	0.769	0.750	0.193
XGBoost	0.833	0.833	0.944	**0.135**
ExtraTrees	0.750	0.727	0.944	0.166
3D physics	0.750	0.667	0.694	0.208
MotionBERT emb. + logit	0.667	0.600	0.833	0.193
LECD-C	**0.917**	**0.909**	**0.972**	0.158

TABLE III. LOCKED SOURCE-DISJOINT TEST CONFUSION MATRIX AT THRESHOLD 0.5.

Actual / pred.	*Incomplete*	*Complete*
Incomplete	6	0
Complete	1	5

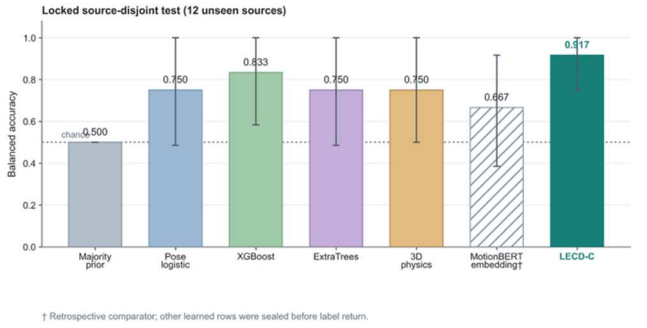


Fig. 5. Locked source-disjoint prospective-test balanced-accuracy comparison on 12 clips.

Fusion-weight sensitivity and branch ablation

Fig.6 reports the development-only sensitivity analysis supporting the selected 2D weight. The pre-specified $\alpha = 0.8$ rule gives the best evaluated mean balanced accuracy (0.788) and event consensus (0.800). Fusion improves consensus balanced accuracy over 2D by 0.053, 3D physics by 0.032, and the frozen embedding by 0.136. Lower 2D weights give slightly better Brier scores, so the claim is limited to the primary balanced-accuracy criterion.

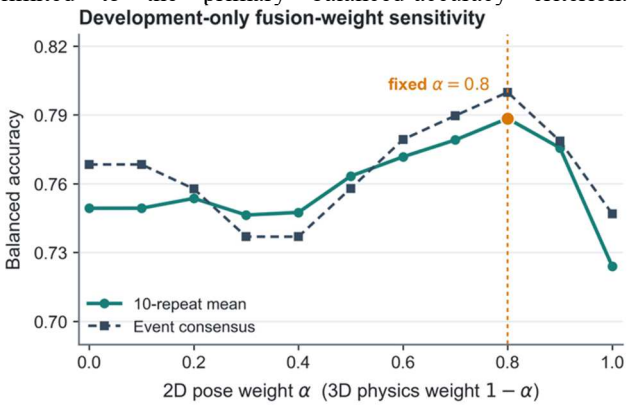


Fig. 6. Development-only fusion-weight sensitivity.

B. Ordinal quality assessment

Table IV reports development-only quality results. LECD-Q obtains 0.431 Macro-F1, 0.436 balanced accuracy, 0.293 QWK, 0.630 tier MAE, and 1.364 score MAE. It improves Macro-F1 over ExtraTrees by 0.017, but ExtraTrees has lower tier MAE and XGBoost ties score MAE. The evidence supports a broad-tier signal, highlighting a research direction of assessing Ballet rotations quality.

TABLE IV. THREE-TIER QUALITY DEVELOPMENT ON 43 EVENTS; 12 × 5-FOLD GROUPED MEANS

Model	*Macro-F1*	*Bal. acc.*	*QWK*	*Score MAE*
Majority prior	0.204	0.333	0.000	1.456
Pose multinomial	0.322	0.324	0.087	1.523
XGBoost	0.287	0.338	0.064	1.364
ExtraTrees	0.414	0.420	0.250	–
LECD-Q	**0.431**	**0.436**	**0.293**	**1.364**

VIII. LIMITATIONS

Several limitations remain and motivate important directions for future research:

- LECD currently focuses on ballet rotations. Extending the framework to jumps, balances, and more diverse dance movements could expand its generality and broaden its utility for dancers and instructors.
- LECD-Q achieves a Macro-F1 of 0.431, indicating that broad quality tiers are learnable. Future work is still required to expand low-score and boundary cases, calibrate multiple expert raters, and investigate ordinal losses, score distributions, technique-level supervision, and 3D temporal features.
- Monocular lifting remains sensitive to occlusion, front-back ambiguity, camera motion, and active acquisition has not yet been compared with passive sampling under equal annotation budgets. In the future, we should incorporate camera-motion compensation, multi-view or inertial observations, and controlled comparisons with random and stratified sampling.

IX. Conclusions

Overall, this study demonstrates the feasibility of reliable and selective ballet rotation assessment for unconstrained monocular video. The main conclusions are summarized as follows:

- LECD, as a selective quality assessment, verifies rotation completion, eligibility, and visibility before evaluating movement quality, thereby preventing incomplete, invalid, or non-scorable videos from being incorrectly interpreted as low-quality performances.
- By combining observable 2D pose dynamics with a frozen 3D rotation prior, LECD-C achieved a balanced accuracy of 0.917 in a label-blind prospective test comprising 12 previously unseen sources, outperforming all evaluated baselines.
- LECD-Q achieved a Macro-F1 of 0.431, the highest result among the quality baselines that had been evaluated. This result indicates that broad ballet-turn quality levels contain a meaningful and learnable signal.
- LECD provides a more reliable foundation for interpretable and evidence-based feedback that may support dancers' technical improvement by establishing a unique, professional, intuitive expert-annotation process which brings humanity into the model and makes the model more reliable as a dancing helper, considering artistic expression or the specific conditions of each rotation.

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